



Fire–climate–human interactions during the Holocene: insights from sediment records in southwestern Türkiye

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Abstract

Fire is a major driver of Mediterranean ecosystems, yet the long-term interactions between fire, vegetation dynamics, and human land use in southwestern Türkiye remain poorly understood. Unlike previous Mediterranean studies, which have tended to focus on individual records or assume direct links between vegetation and fire, this study explicitly compares charcoal, pollen, and anthropogenic indicators. This allows the relative roles of climate, vegetation structure, and human activity in shaping fire regimes to be disentangled. In this study, we use a lacustrine sediment record spanning the last 12,000 years to reconstruct Holocene fire history and its environmental controls, employing charcoal, deciduous *Quercus* pollen, cultivated trees pollen (*Olea*, *Castanea*, and *Pistacia*), and anthropogenic pollen indicators (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*). Charcoal concentrations show some variability over time, reflecting changes in fire activity and discrete fire episodes associated with forest growth and increasing human influence. However, these variations do not consistently align with changes in vegetation, suggesting that fire regimes were influenced by several interacting factors rather than one dominant driver. During the Early Holocene, low charcoal influx and low percentages of deciduous *Quercus* pollen suggest weak, climate-driven fire regimes with limited fuel availability. After 9000 years BP, the rapid expansion of oak-dominated forests led to sustained increases in fire activity and fire episode frequency, peaking during the Middle Holocene between 8500 and 4500 years BP. The close correspondence between high *Quercus* abundances and charcoal peaks highlights the primary role of forest structure in regulating fire regimes. From around 7500 years BP onwards, increasing anthropogenic pollen indicators (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) and cultivated tree (*Castanea*, *Olea*, and *Pistacia*) pollen demonstrate the emergence of arboriculture and managed landscapes, indicating growing human influence on fire activity. During the Late Holocene, declining charcoal accumulation, moderate oak percentages, and the persistence of cultivated trees reflect a transition towards controlled burning and stable agricultural systems. These results demonstrate a progressive shift from predominantly climate- and vegetation-controlled fire regimes in the Early Holocene to increasingly human-modified systems after 7500 years BP, culminating in fully established, managed fire regimes with reduced large-scale burning during the Late Holocene (after 3000 years BP).

Keywords Fire regime · Anthropogenic · Pollen · Charcoal · Eastern Mediterranean

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1 Introduction

Fires have occurred on Earth for at least 420 Ma and are a major driver of many Earth system processes (He and Lamont 2018). The relationship between humans and fire dates back to the Early Pleistocene, with evidence of controlled or habitual fire use by hominins emerging as early as 1.6 Ma ago in eastern Africa (Hlubik et al. 2017, 2019), and additional evidence from southern Africa. On a global scale, humans have been described as a keystone species with regard to fire. They have fundamentally altered fire regimes through their practices of ignition, fuel manipulation, and land-use strategies across different cultural and environmental settings (Bowman et al. 2011). Recent cave excavations along the southern coast of Africa suggest that hominins used fire to transform rocks as early as 70,000 years ago (Clark 2021). Humans modify natural patterns of biomass burning by managing and manipulating vegetation communities, including fire suppression or promotion, and by accidentally or deliberately igniting natural vegetation (Turner et al. 2008). Further archaeological and palaeoecological studies demonstrate that human occupation has played a critical role in shaping ecosystem dynamics through fire. This has contributed to biodiversity loss, changes in vegetation, and landscape transformation (Gosling et al. 2017, 2020). Recent archaeological and palaeoecological studies demonstrate that the region has experienced substantial and long-term human influence, including active forest management, biodiversity modification, and controlled fire use, over millennial timescales. These findings emphasize that Amazonian landscapes are largely the result of ongoing interactions between humans and the environment rather than being pristine natural systems (Gosling et al. 2021; Furquim et al. 2023). Fire regimes reflect the interaction between climate, vegetation, and human activity. Global charcoal syntheses reveal climate-dominated patterns since the Last Glacial Maximum, accompanied by increasing variability during the Holocene (Power et al. 2008; 2010). However, recent studies emphasize the growing influence of humans, particularly during the Late Holocene, with significant regional variations (Bremond et al. 2024). Wildfires have become more frequent in recent years due to droughts, heatwaves, and strong winds associated with climate change (Heibati et al. 2025). Wildfires are one of the most significant agents of ecological disturbance, affecting vegetation cover, soil, and biodiversity on decadal to centennial timescales (Vannière et al. 2008), including the global distribution of tree cover (Yigit 2025). More generally, wildfires help maintain open grasslands by promoting their growth and woody encroachment (Turner et al. 2010a, b). Wildfires also pose one of the greatest threats to pine ecosystem sustainability in the Mediterranean Basin (Enriquez de

Salamanca 2023). Therefore, the historical fire regimes of fire-adapted Mediterranean pine forests are of great interest from ecological and social perspectives (Fulé et al. 2008). Fire regimes are influenced by a range of factors, including regional controls such as climate (temperature, precipitation, and wind), fuel characteristics (load, connectivity, and flammability), ignition regimes (seasonality and location, whether natural or anthropogenic), and local landscape factors (soil type, slope, and exposure) (Vannière et al. 2016).

The role of fire in ecosystem dynamics is increasingly important, given that projected climate warming is likely to influence fire regimes (Vannière et al. 2008). Fire regimes, defined by severity (an indicator of the amount of biomass burned), frequency (the number of events over time), and extent, in forested areas of the Mediterranean region and elsewhere, are closely linked to hydrologic conditions and seasonality, which, in turn, are related to large-scale atmospheric circulation patterns (Vannière et al. 2008). The lasting and pervasive influence of fire on terrestrial ecosystems has generated tremendous scientific interest in understanding its environmental drivers and effects. Nevertheless, the impact of human-induced fires on fire regimes, biomass burning, and fire-related carbon emissions at regional to continental scales remains largely unresolved (Vannière et al. 2016).

One approach in identifying the past drivers of fire regimes at the regional scale is to identify key periods characterized by homogeneous fire patterns and to link these periods to broad-scale environmental or anthropogenic factors (Vannière et al. 2016). Microscopic charcoal, which can be transported over long distances (20–100 km), is used as a proxy for regional fire activity (Brechtbühl et al. 2024). The analysis of charcoal fragments from sedimentary materials is a common method in paleoecological research for reconstructing aspects of past fires, particularly fire frequency (Nascimento et al. 2023). Recurrent fires have been shown to promote herbaceous species in Mediterranean-type ecosystems, particularly in areas where short fire intervals occur (Beatriz and Ramón 2008). Furthermore, wood charcoal assemblages from archaeological sites can be used to reconstruct woodland communities that no longer exist and reveal how past human societies procured and used wood resources (Marston 2012; Vachula et al. 2021; Thapa et al. 2022). Microscopic charcoal can be transported over distances of tens to hundreds of kilometers and is therefore widely used as a proxy for regional fire activity (Leys et al. 2015, 2017; Vachula et al. 2018). However, depending on particle size, transport processes, and depositional context, charcoal records reflect a combination of local and regional fire signals (Graves et al. 2019). Larger particles generally indicate local fire events, whereas smaller particles may represent broader regional burning (Clarke and Royalle 1996;

Whitlock and Larsen 2001). In the last decades, Mediterranean fire regimes have been investigated using a variety of approaches. At a global scale, synthesis and modeling studies emphasize the interactions between climate, vegetation, and fire, with a particular focus on the roles of temperature, moisture balance, and fuel availability (Vannière et al. 2008; Pausas and Fernández-Muñoz 2012; Kelly et al. 2013; Barrett et al. 2013; Zhao et al. 2025). At the proxy-based level, sedimentary charcoal analysis is the primary tool for reconstructing past fire activity, providing long-term records of fire frequency and magnitude. In the western Mediterranean, charcoal studies, often integrated with archaeological data, have been widely used to explore palaeofire dynamics and human–environment interactions (Carrión et al. 2003; Figueiral and Carcaillet 2005; Gil-Romera et al. 2010; Curt et al. 2013; Leys et al. 2024). In the eastern Mediterranean, multi-proxy approaches combining charcoal, pollen and archaeological evidence have been used to understand the influence of climate and land use on fire regimes, particularly in Greece (Christopoulou et al. 2013; Cañellas-Bolta et al. 2018) and Anatolia (Wick et al. 2003; Turner et al. 2008, 2010a, b; Asouti and Kabukcu 2014; Şahan et al. 2021, 2022, 2023; Brechbuehl et al. 2024; Eker et al. 2024). More recent studies integrate charcoal with geochemical proxies and statistical approaches to better distinguish between climatic and anthropogenic influences (Badeau et al. 2024; Poutli et al. 2024). Despite these advances, distinguishing between climate and human influences remains challenging, particularly in Anatolia, where high-resolution, well-dated multi-proxy records are scarce. This study, therefore, uses sedimentary charcoal, supported by pollen and complementary proxies, to reconstruct Holocene fire regime dynamics in southwestern Anatolia and evaluate the relative roles of climate and human activity. The topography, diverse flora and fauna, and a long history of human habitation have given rise to various microenvironments with high biodiversity, attracting successive civilizations in southwestern Türkiye (Eastwood et al. 1999; Médail and Quezel 1999; Bakker et al. 2012). Forest fires are a major ecological force shaping and transforming forested areas in the region (Eker et al. 2024). Eceler Höyük (mound, southwestern Anatolia; Fig. 1) lies within the Mediterranean climatic and vegetational zone. Despite its importance from both natural and human historical perspectives, it remains relatively understudied, with few archaeological and geomorphological investigations to date (Aksoy and Köse 2005; Momigliano et al. 2011). Reconstructing past fire regimes provides a long-term baseline against which to distinguish natural variability from recent changes in fire activity. This offers valuable context for improving regional fire management and risk assessment strategies. Following Keeley (2009), the fire-related terminology was used in this study. 'Fire activity'

refers to the total amount of biomass burned, as determined by charcoal accumulation. 'Fire episodes', on the other hand, refer to statistically significant peak events identified in the residual charcoal series. The term 'fire regime' describes the combined characteristics of fire frequency, magnitude, and temporal variability. Direct inference of fire activity from charcoal data is avoided, since sedimentary charcoal primarily reflects variations in biomass burning rather than combustion intensity.

This study presents a sub-millennial sedimentary record of regional fire history from the Younger Dryas to the present at the Seki-Eceler Höyük archaeological site in southwestern Turkey. This record is then correlated with records from central Anatolia and the surrounding regions, including Greece and the Caucasus.

2 Study area

Seki-Eceler, southwestern Türkiye (Muğla province; 36° 48' 52.1" N 29° 38' 13.1" E, Fig. 1), is an eastern part of the Lycian ophiolitic melange (Danelian et al. 2006), with an elevation of 1100 m, overlain from top to bottom by the Lycian thrust sheets, or the Lycian Nappes. The Lycian Nappes are located on a vast autochthonous basement beneath the Menderes Massif in the northwest and the Beydağları Platform in the southeast (Collins and Robertson 1998; Danelian et al. 2006; Alçiçek et al. 2015; Keserci et al. 2023). This basin is surrounded by Akdağlar in the west and Boncuk Dağları to the northeast (Aksoy and Köse 2005). The Eceler Höyük on the Seki Plateau covers an area of 33 ha and reaches a height of 20–25 m (Fig. 2). The oldest artifact discovered here dates to the Late Neolithic and Early Chalcolithic (7500 years BP) (Aksoy and Köse 2005; Momigliano 2013).

Seki Basin, as a part of southwestern Anatolia, is characterized by a Mediterranean climate and vegetation (Özhatay et al. 2013; Tüfekcioğlu and Tavşanoğlu 2024; Bağçacı et al. 2024). The most well-known pressure patterns affecting the Mediterranean region are the North Atlantic Oscillation (NAO), the East Atlantic (EA) pattern, the East Atlantic–Western Russian (EA–WR) pattern, and the Mediterranean Oscillation (MO) (Cenk and Turgay 2020; Criado-Aldeanueva and Soto-Navarro 2020). NAO is the main oscillation that affects the Mediterranean region. A stronger NAO is associated with a cooler, drier climate, and a weaker NAO with a warmer, wetter climate (Danladi et al. 2018; Lionello et al. 2023). The Seki Plateau, at an altitude of 1100 m above sea level, with an annual rainfall of 571.2 mm and an annual temperature of 12.7 °C (Bozyiğit 1997), has a relatively cool climate in which snow remains on the ground in winter and rainfall decreases in spring. Rainfall on the Seki

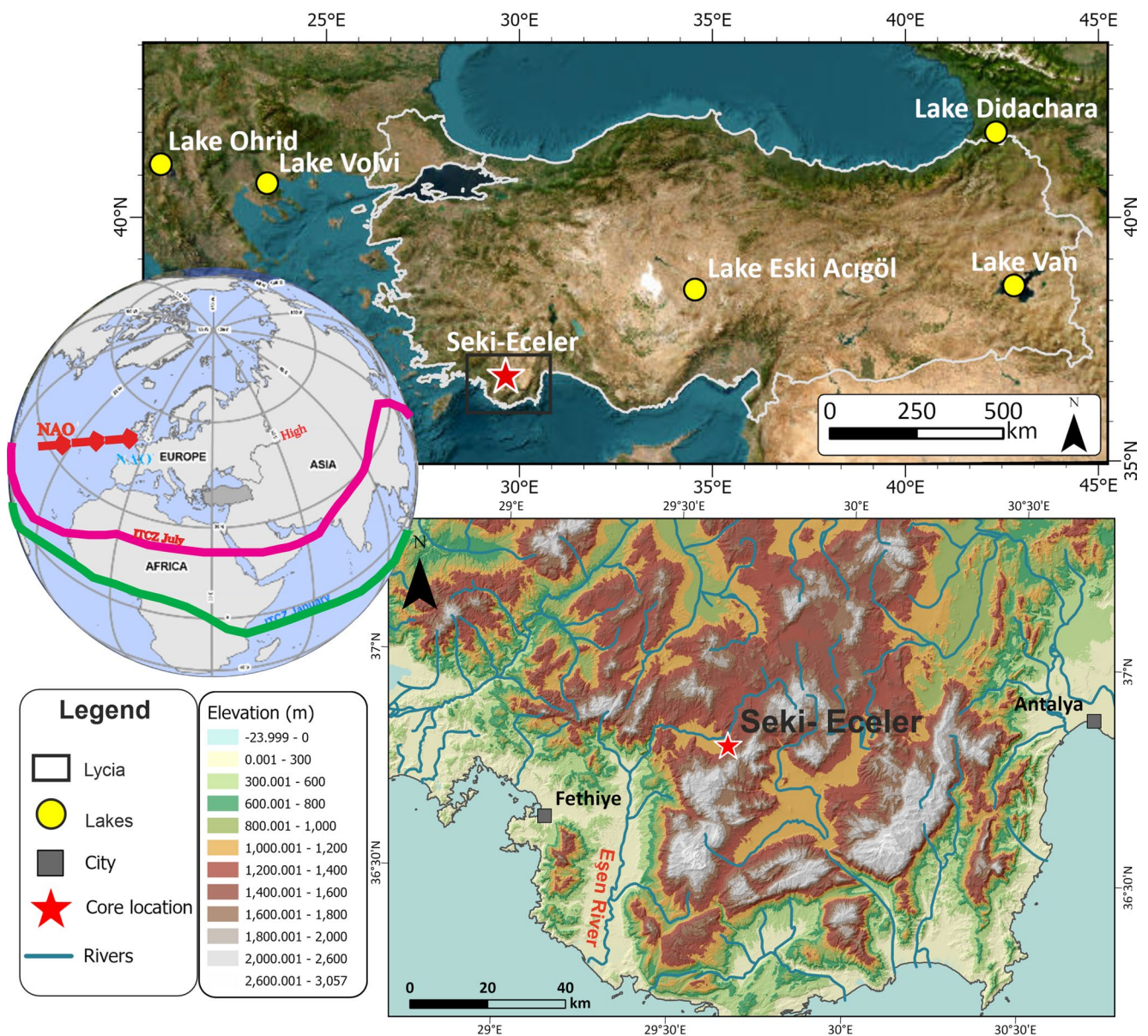


Fig. 1 Location map of the Seki-Eceler Basin and compared areas. The core was taken from the Seki-Eceler Basin. Charcoal data of Seki-Eceler with charcoal data from Lake Ohrid and Lake Volvi (located in the Balkans), Lake Didachara (Caucasus), Acıgöl, and Lake Van (Anatolian region of Türkiye). The inset map illustrates the broader climatic

setting, highlighting the influence of large-scale atmospheric systems such as the North Atlantic Oscillation (NAO) and the seasonal migration of the Intertropical Convergence Zone (ITCZ). The base map is derived from satellite imagery, and the scale bar represents 500 km

Plateau is lower than in the coastal regions. The Mediterranean vegetation ranges from deciduous forests to summer-dry scrub and woodland. The natural vegetation type around the Eşen Çayı catchment is maquis (a dense Mediterranean shrubland biome dominated by evergreen, sclerophyllous shrubs and small trees) up to an elevation of 800 m. *Abies cilicaca* sp. and *Quercus* grow in higher elevations than 800 m, and *Temble*, *poplar*, *eastern Strawberry*, and *Cistus europeus* exist in the highest land (Bozyiğit 1997).

The Seki Plateau is the first basin that the settlers of Fethiye reached and used as a summer pasture (Lycian

yayla) (Momigliano et al. 2011). The mounds on this plateau, such as Eceler, represent the most prominent settlements and contain a wealth of yet-undiscovered prehistoric pottery that can provide valuable insights into the Late Neolithic/Early Chalcolithic, Early Bronze Age, Iron Age, and Roman periods (Momigliano 2013). This evidence demonstrates that the höyük has a long, continuous settlement history dating back to the Late Neolithic (7500 years BP) (Aksoy and Köse 2005; Momigliano et al. 2011; İşkan and Dündar 2016). Moreover, despite its high altitude, the plateau functioned as an important transportation corridor

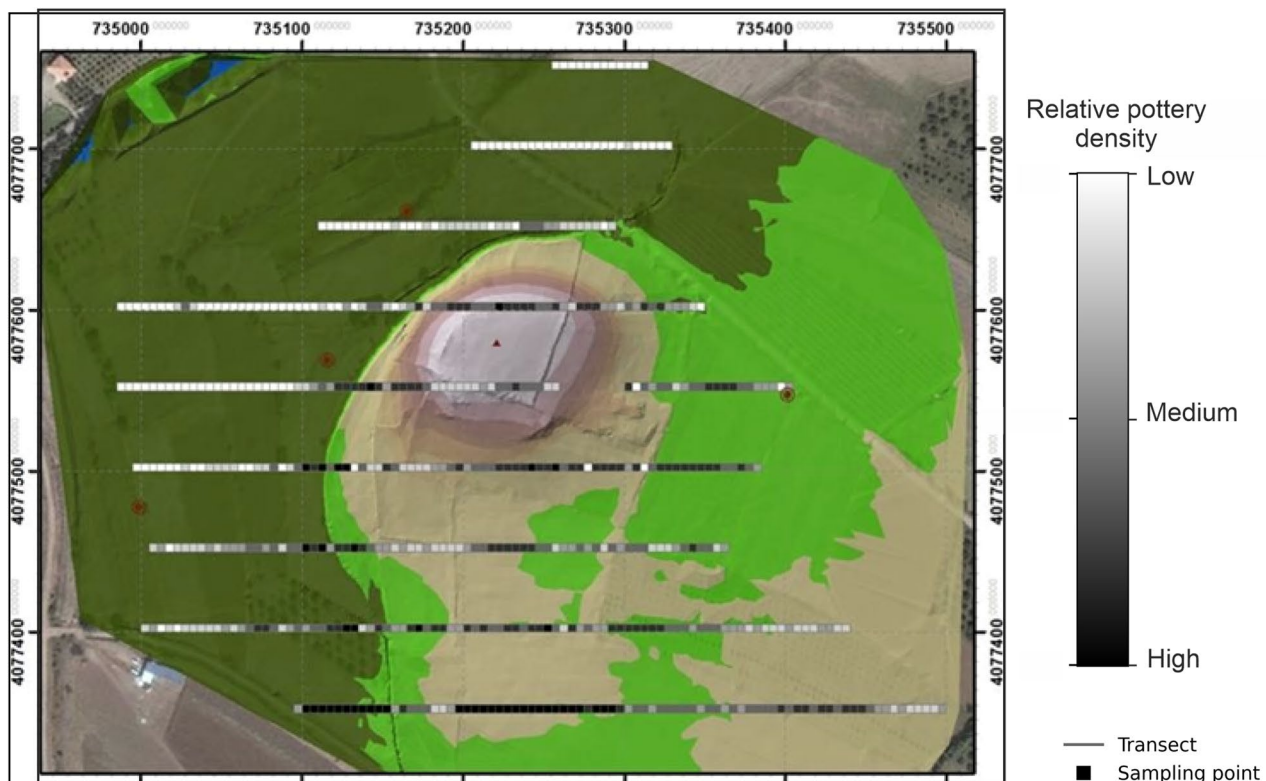


Fig. 2 Digital elevation model (3D) of Seki-Eceler Höyük superimposed on a satellite image, showing pottery density distribution along systematic transects across the mound. Pottery density is represented using a grayscale gradient (light=low density; dark=high density), with values corresponding to sherd counts per sampling unit. Horizontal

transects indicate sampling lines, and squares along transects represent individual sampling locations. The central mound corresponds to the highest artifact concentration, with decreasing densities toward the periphery. A total of 9000 ceramic sherds were collected

between Anatolia and the Aegean, indicating that it was not merely a seasonal occupation area but a year-round habitation zone (Momigliano et al. 2011).

3 Materials and methods

3.1 Coring, sampling and sedimentological studies

The JARK14-04 sediment core was recovered via a hydraulic vehicle-mounted rig adjacent to Seki-Eceler Höyük in southwestern Türkiye, in 2014. Micro-X-ray fluorescence (μ XRF) and Total organic carbon (TOC) was performed on the core at the time of coring (2014). A total of 42 samples were taken at 10 cm intervals along the entire length of the core at the Istanbul Technical University EMCOL Applied Research Centre. The lithological characteristics of the core were systematically identified and recorded, and the stratigraphic sequence was documented using vertical logging.

3.2 Chronology and age-depth modelling

Core JARK14-04 chronology is based on ten AMS ^{14}C dates obtained from bulk-sediment samples (Poznań Radiocarbon Laboratory; Table 1). Radiocarbon determinations were calibrated using IntCal20 (Reimer et al. 2020) and an age–depth model was constructed in rbacon (Blaauw and Christen 2011) in R. The model was additionally anchored by a core-top calendar-age tie point of -60 ± 10 cal years BP at 0 cm (relative to AD 1950).

Because bulk sediment can incorporate variably old (reworked/dead) carbon, one stratigraphically inconsistent determination was treated as a potential outlier and down-weighted using a heavy-tailed Student-t likelihood, whereas the remaining dates were assigned near-normal likelihoods. To account for an interval of rapid sedimentation/event deposition between 130 and 197 cm, this depth range was specified as a slump/event layer in Bacon, producing an abrupt step in the age–depth relationship across the shaded interval (Fig. 3). Sedimentation priors were otherwise kept at default settings, but a relatively strong memory prior was used to reflect persistent accumulation expected for fine lacustrine sediments (mem.mean=0.8; mem.strength=20).

Table 1 AMS radiocarbon and calibrated ages of core JARK14-04. All measurements are on bulk sediment total organic carbon (TOC). Ages are reported as radiocarbon years BP (relative to AD 1950) $\pm 1\sigma$. The last column lists the per-date Student-t likelihood hyperparameters (t.a, t.b) used in rbacon to reflect differing reliability: (33, 34) approximates a near-normal likelihood, whereas (3, 4) is heavy-tailed to down-weight potential outliers. No per-sample calibrated ages are given here; calibration is performed only within the Bayesian age-depth model (see Methods). A calendar-age tie point at the core top (0 cm, -60 cal years BP, $\sigma=10$ years) was applied separately and is not included in this table

Lab code	Dated material	Core depth (cm)	^{14}C years BP	(t.a, t.b)
Poz-170065	TOC	75	3125 $\pm$ 35	(33, 34)
Poz-170066	TOC	102	4210 $\pm$ 30	(3, 4)
Poz-169999	TOC	197	4210 $\pm$ 30	(33, 34)
Poz-95796	TOC	210	4005 $\pm$ 35	(3, 4)
Poz-95794	TOC	250	6460 $\pm$ 40	(33, 34)
Poz-170067	TOC	309	2300 $\pm$ 30	(3, 4)
Poz-95768	TOC	340	7430 $\pm$ 50	(33, 34)
Poz-170228	TOC	377	9390 $\pm$ 60	(33, 34)
Poz-170095	TOC	418	9660 $\pm$ 60	(33, 34)
Poz-95767	TOC	450	10,050 $\pm$ 70	(33, 34)

Posterior ages are reported as medians with 89% highest

posterior density (HPD) intervals, and ages below the deepest dated level (450 cm) were extrapolated using Bacon's prior accumulation structure (Fig. 3).

3.3 Charcoal analysis and counting

For pollen analysis, sediment samples were collected at 10 cm intervals and processed according to standard protocols (Cour 1974). This included using cold HCl and HF to remove carbonates and silicates, respectively. Zinc chloride (ZnCl_2) was then used to separate palynomorphs (density > 2), after which the samples were sieved through 200 μm and 10 μm meshes. The pollen residues were mounted in glycerine and analyzed via transmission light microscopy at 63 $\times$. Microcharcoal particles were counted on the same slides that were used for pollen analysis. Only particles larger than 20 μm were included in the counts, and this threshold was applied consistently to all samples to ensure the comparability of the dataset. Charcoal particles were counted on a single slide per sample, and analytical replication was not systematically performed. Consequently, no standard deviations derived from replicate measurements are available. Charcoal counts were

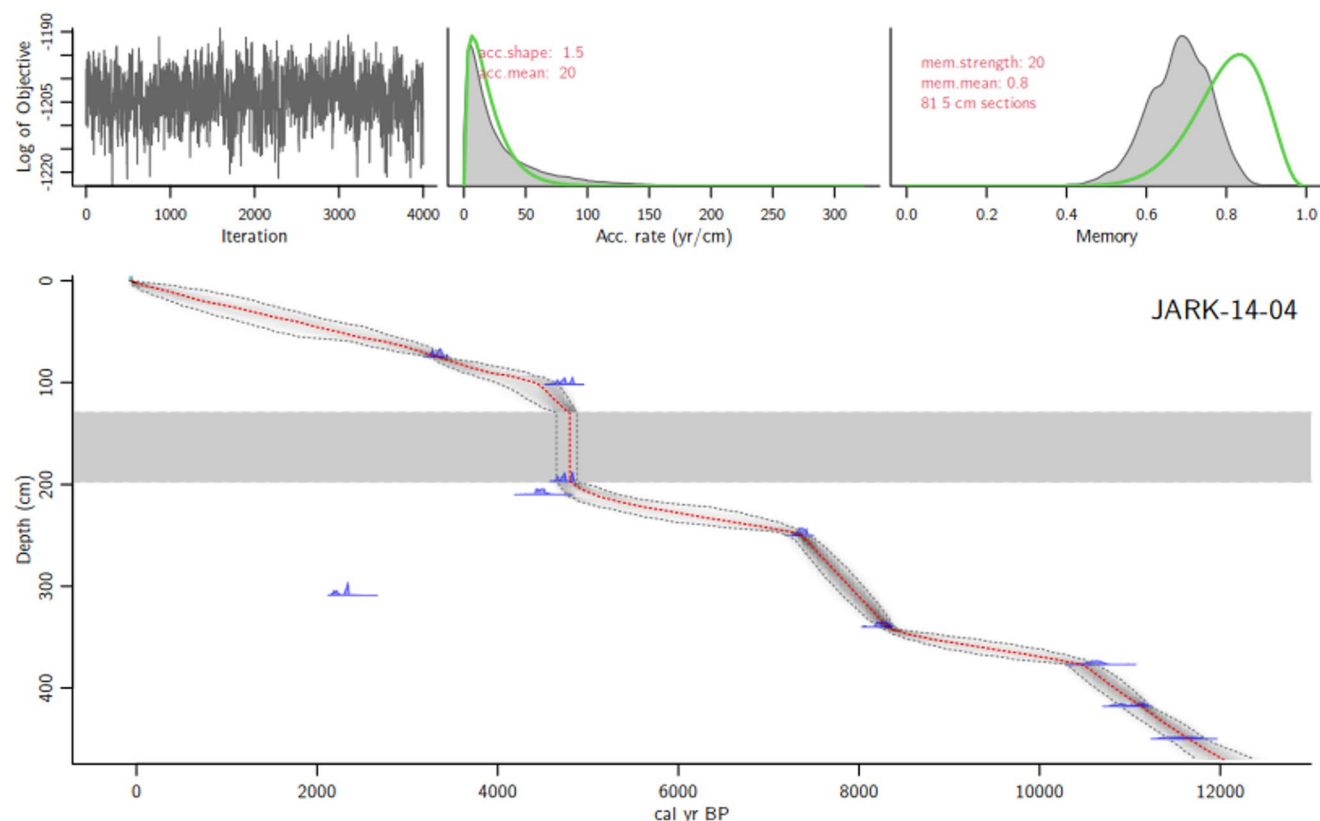


Fig. 3 Age-depth model for the Seki-Eceler core (JARK14-04) constructed with rbacon. The red dashed line shows the posterior median age-depth relationship and the grey envelope indicates the 89% credible interval. The grey shaded band marks the slump/rapid-accumu-

lation interval (130–197 cm) implemented in the model. Top-left: MCMC trace plot. Top-middle and top-right: prior (green) versus posterior (grey) distributions for the accumulation rate (gamma prior) and memory (beta prior), respectively

converted to concentration values and used to reconstruct fire activity through time. To facilitate comparison between samples with different absolute values, charcoal data were also standardized using a *z-score* transformation:

$$Z = \frac{(X - \mu)}{\sigma}$$

where X is the charcoal value, μ is the mean, and σ is the standard deviation of the record.

To distinguish significant fire events from background variability, charcoal peak detection was performed on the residual series obtained after smoothing (Higuera et al. 2009). Peaks exceeding the 95th percentile threshold of the residual distribution were considered statistically significant and interpreted as discrete fire episodes representing periods of elevated fire activity. In this framework, background charcoal (BCHAR) is used as an indicator of long-term fire activity, while peak charcoal (PCHAR) represents individual fires. Because charcoal data do not allow direct quantification of fire activity, interpretations are limited to changes in fire activity and frequency. Charcoal concentrations were log₁₀-transformed prior to analysis. Background charcoal (BCHAR), representing regional fire activity and long-term biomass-burning trends, was estimated using LOWESS smoothing (Locally Weighted Smoothing) with a smoothing span of 0.25. Peak charcoal (PCHAR), interpreted as local fire events, was calculated by subtracting the background component from the log-transformed charcoal series. Fire events were defined strictly as peak charcoal values exceeding the 95th percentile of the residual (PCHAR) series. Peaks below this threshold were not considered fire episodes, even if visually distinct from the background curve. In addition to charcoal analysis, selected pollen taxa were used to derive anthropogenic pollen indicators (Hashemi et al. 2026). These indicators are based entirely on pollen data and include anthropogenic pollen (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*), cultivated tree pollen (*Castanea*, *Olea*, and *Pistacia*), and percentages of deciduous *Quercus*.

4 Results

4.1 Chronology and lithology

According to the age–depth model, the 4.7 m long JARK14-04 core provides a well-resolved chronological framework for sediments from the Seki–Eceler Höyük area. The Bayesian rbacon model accommodates one clearly inconsistent radiocarbon determination by down-weighting it with an outlier-robust likelihood (Fig. 3). The resulting age–depth

relationship indicates that sediment accumulation was not constant through time. In particular, the model shows an abrupt step across 130–197 cm, consistent with a rapid-deposition/event (slump) interval rather than steady background sedimentation. Outside this interval, accumulation varies more gradually with depth, and deeper ages are constrained by the dated levels and the model priors. The lower part of the core, dated to approximately 12,000 years BP, corresponds to the Younger Dryas interval and consists of brown, clayey, sandy silt. This unit represents the Younger Dryas–early Holocene transition at a depth of 450 cm (Fig. 4). Above this, the succession continues with reddish-brown, sandy, clayey silt containing root-like structures from 310 to 220 cm (7990–5530 years BP). The silt layer was deposited during the transition from the Early to the Middle Holocene. This unit is overlain by reddish-dark brown, sandy, clayey silt enriched with carbonate pebbles. A thin layer of reddish-brown, clayey, silty fine sand containing distinctive orange mottles (1–8 mm) caps this interval, and its upper boundary is dated to 4780 years BP at 197 cm. Between 197 and 130 cm, the core consists of dark brown, clayey, silty fine sand containing black reduction spots and black chert or obsidian pebbles, as well as discrete horizons of flat grey ceramics (Fig. 4). Ceramic pieces also were documented at 125 cm (4700 years BP). The top of this unit, at 130 cm, is dated to 4780 years BP. The uppermost part of the sequence comprises a layer of dark brown, sandy, clayey silt with rare subrounded pebbles, 130 cm thick. The uppermost dated sample, taken at a depth of 75 cm, yields an age of approximately 3330 years BP. The topmost part of the core lacks a direct chronological range and is extrapolated from the first radiocarbon age. Thus, the chronology for this interval should be interpreted with caution. The average sediment accumulation rate is estimated at ~0.5 mm per year, corresponding to approximately 20 years per centimeter.

4.2 Charcoal analysis

The charcoal record, as revealed by palynological analysis, shows significant temporal variability throughout the Holocene (Figs. 5 and 6). The range of charcoal concentrations is from 157 to 6696 particles per sample. Most values cluster between 600 and 2000 particles, indicating substantial variability over time. Charcoal concentrations, vegetation composition, and anthropogenic pollen indicators (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) show significant and consistent temporal variation from the terminal Pleistocene to the present day (Figs. 5 and 6). During the terminal Younger Dryas and the earliest Holocene (approximately 12,000–11,000 years BP), log-transformed charcoal values are generally low, ranging from ~2.5 to 2.7. Fire peaks are rare, and total charcoal counts

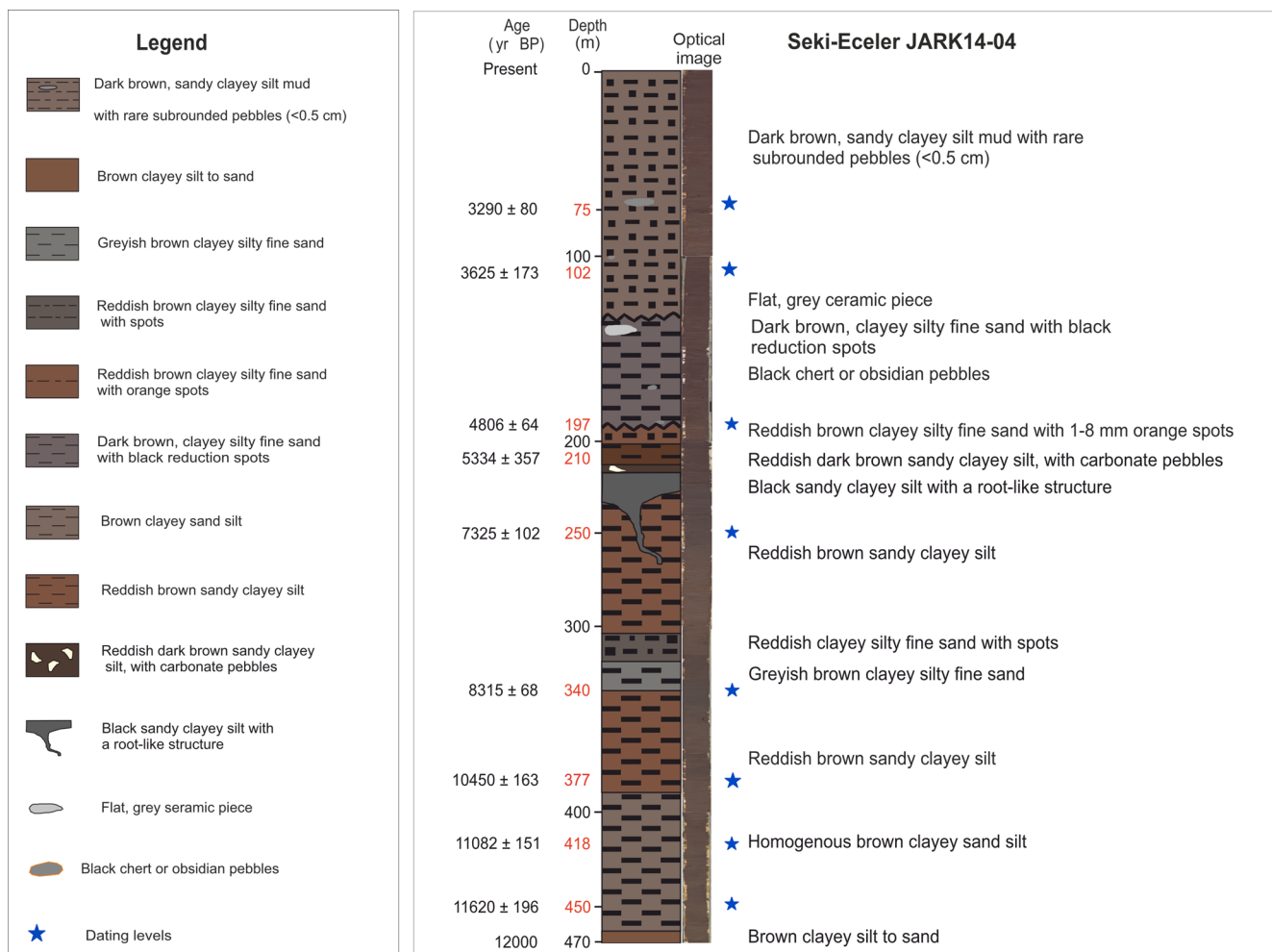


Fig. 4 Lithology of the core JARK14-04 with optical image of the core. Stars showed the points in chronological order. The calibrated ages of each sample were written in the left side of the photograph. Red numbers were the depths that were sampled for the dating

mostly remain below 4000–5000. During this period, deciduous *Quercus* percentages were generally below 20–30%, and anthropogenic pollen indicators (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) remained close to background levels. From around 11,000 to 9000 years BP, charcoal concentration gradually increased, with background values rising to approximately 2.8–3.0, and several low- to moderate-amplitude fire peaks appearing. The smoothed charcoal curve shows a progressive upward trend. This increase coincides with a gradual rise in the percentage of *Quercus* pollen, whereas indicators of human activity and cultivated tree (*Castanea*, *Olea*, and *Pistacia*) pollen remain absent or extremely rare.

After 4500 years BP, charcoal concentrations remained elevated, but became more variable. Peak magnitudes decreased slightly, although repeated fire events persisted between 4000 and 3000 years BP. At the same time, the percentage of *Quercus* began to decline after ca. 4500–4000 years BP, reaching moderate values of 20–40%.

Meanwhile, anthropogenic pollen indicators (Asteraceae, Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) remained high, while cultivated tree pollen (*Castanea*, *Olea*, and *Pistacia*) became more frequent and persistent.

From around 3000 years BP to the present day, charcoal accumulation has declined. Background values decreased to 2.3–2.7, total charcoal counts mostly fell below 4000–5000, and fire peaks became less frequent and weaker. Meanwhile, deciduous *Quercus* percentages remain moderate, anthropogenic pollen indicators persist, and cultivated tree pollen (*Castanea*, *Olea*, and *Pistacia*) shows prominent peaks around 3500–3000 years BP and again near 2000 years BP. This interval coincides with the forest's maximum development, with deciduous *Quercus* reaching peak abundances of 60–90%. At the same time, anthropogenic pollen indicators (Asteraceae, Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) show a clear increase after 8000 years BP, with repeated peaks between 7500

and 4500 years BP, reaching values of ~700–1,000 units. Cultivated tree pollen (*Castanea*, *Olea*, and *Pistacia*) first appears around 12,000 years BP, occurring relatively frequently but at low levels compared to other pollen taxa. It becomes more consistently present during the Middle Holocene, notably around 8500–8000, 7200, and 3400 years BP.

Identifying statistically significant fire events is further refined by charcoal peak detection based on the residual series. Using a 95th percentile threshold, two distinct peaks were detected at 9736 and 4714 years BP. These peaks represent periods when charcoal accumulation exceeds background variability, indicating discrete fire episodes. The earliest peak, at 9736 years BP, occurs during the Early Holocene. The most prominent peak is observed at 4714 years BP, corresponding to the highest charcoal values in the sequence.

After *ca.* 5000 years BP, charcoal concentrations remain relatively high, but become more variable. Peak magnitudes also decrease slightly, though repeated fire events persist between 4000 and 3000 years BP. Concurrently, deciduous *Quercus* percentages begin to decline after *ca.* 4500–4000 years BP, reaching moderate values of ~20–40%. Meanwhile, anthropogenic pollen indicators remain consistently high, and cultivated tree pollen becomes more frequent and persistent. Despite numerous visually prominent peaks in the charcoal record, only three exceed the statistical threshold. This indicates that major fire episodes were infrequent and episodic, whereas background variability reflects continuous fire activity.

5 Discussions

5.1 Holocene fire regimes in relation to climate, vegetation dynamics, and human activity

The charcoal record sheds light on past fire activity and enables us to assess how fire regimes responded to long-term changes in climate, vegetation structure, and human land use during the Holocene (Figs. 5 and 6).

5.1.1 Early holocene (*ca.* 12,000–8500 years BP): Climate-driven fire regimes during postglacial forest expansion

During the earliest Holocene (*ca.* 12,000–11000 years BP), charcoal concentrations were very low. Log-transformed background values were generally between 2.6 and 2.8, and total charcoal counts were mostly below 4000. Fire activity was weak and infrequent, as indicated by the sparse, low-amplitude peaks. This pattern reflects the environmental conditions that prevailed after the Younger Dryas period, when limited vegetation cover and low biomass

availability restricted the spread of fires despite rising temperatures. Central Anatolia including Eski Acıgöl (Turner et al. 2008), and Lake Van (Wick et al. 2003), and some places in the North Macedonia such as Lake Ohrid (Brechtbühl et al. 2024), and in the Greece, for instance, and Lake Volvi (Ganz et al. 2025) had low fire events, as well (Fig. 7). The standardised charcoal comparison confirms that the majority of records remain below the 95th percentile threshold during this period, suggesting an absence of significant fire peaks and corroborating the hypothesis of generally low fire activity (Fig. 7).

Similar low charcoal influxes during the terminal Pleistocene–early Holocene transition have been reported at Lakes Van, Zeribar, and Nar (Wick et al. 2003; England et al. 2008; Roberts et al. 2016), where limited fuel continuity, rather than a cool climate alone, constrained fire activity. Between 11,000 and 9500 years BP, charcoal concentrations gradually increased, with short-lived peaks and a rising background trend, indicating enhanced fire activity associated with vegetation expansion. However, these increases do not consistently manifest as statistically significant peaks in all records. This suggests that fire activity was episodic and spatially variable rather than coherent on a regional scale. During this transition, the statistically significant charcoal peak at 9736 years BP likely reflects the initial increase in biomass burning associated with postglacial forest establishment. Rising deciduous *Quercus* percentages, although still below 20–30%, indicate the development of open woodlands, where increasing fuel availability enabled more sustained fire activity under warm and seasonally dry conditions.

Notably, a comparison of vegetation and charcoal records suggests that increases in *Quercus* do not systematically coincide with charcoal peaks. This indicates that fuel availability alone does not directly control fire activity. Instead, ignition frequency and short-term climatic variability are more likely to have played a key role. The increase in fire activity during the Late Holocene is evidenced by several records (Lake Van, Diachara, and Lake Volvi). This pattern is consistent with the findings of previous studies, which suggest that fire regimes in the Mediterranean were increasingly influenced by human activity. By contrast, sites such as Seki Eceler and Lake Ohrid exhibit more sporadic fire activity, with fewer, albeit sometimes intense, peaks (Fig. 7). These patterns suggest that fire occurrence in these regions was more strongly influenced by climatic extremes or local fuel availability than by sustained human pressure.

Similar Early Holocene increases in charcoal associated with forest development have been documented at Lake Nar (Roberts et al. 2016), as well as across the eastern Mediterranean and the Balkans (Vannière et al. 2011; Colombaroli et al. 2007). Moreover, the amount of charcoal remained

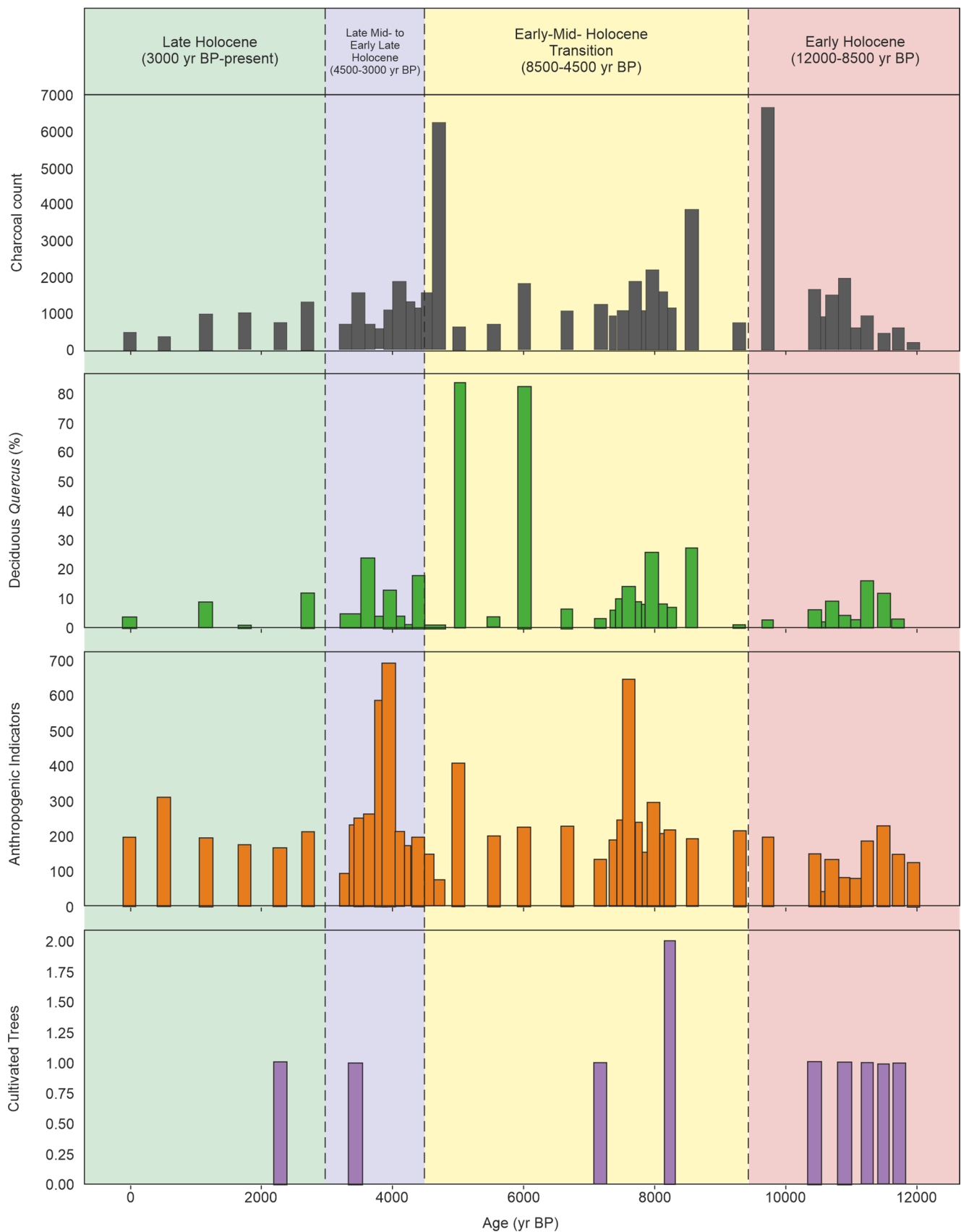


Fig. 5 Plotting stratigraphic variations in total charcoal concentrations (this study) with anthropogenic pollen indicators (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*), cultivated tree pollen (*Castanea*, *Olea*, and *Pistacia*), and deciduous *Quercus* percentages against age (years BP) throughout the Holocene (Hashemi et al. 2026). All data are plotted against calibrated age (years BP). Vertical dashed lines and shaded bands show the main time intervals discussed in the text: Early Holocene (12,000–8500 years BP), Early–Mid Holocene transition and Middle Holocene (8500–4500 years BP), Late Mid-Holocene to Early Late Holocene (4500–3000 years BP), and Late Holocene (3000 years BP–present). The x-axis reflects the age model

constant for Lake Volvi, Didachara, and Lake Eski Acıgöl during the Early Holocene, indicating limited forest expansion in the Caucasus and North Macedonia (Ganz et al. 2025; Connor et al. 2018; Turner et al. 2008) (Fig. 7). However, the standardized comparison shows that the records for lakes such as Volvi, Ohrid, and Eski Acıgöl remained well below the peak threshold during this period. This indicates that fire activity was not elevated uniformly across the region.

Towards the end of the Early Holocene (ca. 9500–8500 years BP), charcoal concentrations decline again, indicating stabilization of fire regimes. This decrease is consistently reflected in the standardized records, where values remain below threshold levels, indicating a reduction in fire frequency. This decrease likely reflects forest maturation, increased canopy closure, and reduced flammability under more humid conditions. Similar post-expansion declines have been reported from Lakes Van and Iznik, where stable forest cover appears to have limited fire frequency once the ecosystems reached a quasi-equilibrium state.

5.1.2 Early–mid holocene transition and middle holocene (ca. 8500–4500 years BP): interaction of forest maturity, climate variability, and emerging human influence

From around 9000 years BP onwards, there have been significant changes in ecosystem functioning and fire regimes. Background charcoal levels and peak frequency both increase markedly, with the smoothed curve rising to ~3.1–3.2, and with several prominent peaks occurring between 8500 and 7500 years BP. Total charcoal concentrations repeatedly approach or exceed 8000–10000 counts, indicating the onset of sustained, intensified fire activity. Regionally, high fluctuations in fire activity were recorded; for instance, Lake Volvi, Didachara, Eski Acıgöl, and Lake Van experienced such fluctuations (Wick et al. 2003; Turner et al. 2008; Connor et al. 2018; Ganz et al. 2025). In contrast, Lake Ohrid in the Balkans showed an increase in fire activity without any fluctuations (Brechtbühl et al. 2024) (Fig. 7).

At the same time, deciduous oaks expanded significantly, reaching a maximum coverage of 60–90% between 6500 and 5000 years BP (Hashemi et al. 2026). This indicates

that dense, oak-dominated forests developed under warm and relatively humid climatic conditions. While the formation of closed forests likely increased the availability and continuity of fuel, the charcoal record does not directly correspond to peaks in deciduous *Quercus*. This suggests that fire activity during this period was influenced by factors other than vegetation density, such as climate variability or human activity. The close correspondence between high oak percentages and elevated charcoal values suggests that forest structure primarily controlled fire regimes during this period, as was also observed at Lakes Beyşehir, Eğirdir, Eski Acıgöl, and other Anatolian sites (Eastwood et al. 1999; Turner et al. 2008; Bottema and Woldring 1984). A significant charcoal maximum around 6000–5500 years BP coincides broadly with the Holocene climatic optimum in the eastern Mediterranean (Jalut et al. 2009). Although this period is generally considered humid, increased seasonality and drier summers likely made biomass-rich landscapes more susceptible to fire. Similar Middle Holocene fire maxima have been reported in southwestern Anatolia and the Levant, suggesting that firing was a regional phenomenon. However, low charcoal levels in Lakes Ohrid and Volvi in the Balkans, and in the Didachara region in the Caucasus, indicate limited forest expansion and a cold climate in the Earth's northern latitudes (Brechtbühl et al. 2024; Ganz et al. 2025) (Fig. 7).

The most significant fire event in the entire record occurs at 4714 years BP and is identified as a peak that exceeds the 95th percentile threshold. This peak coincides with the maximum accumulation of charcoal and represents the culmination of interactions between high biomass availability, favorable climatic conditions, and increasing human influence. Additional high-amplitude peaks around 6.0 and 4.2 ka BP further indicate high-magnitude charcoal peaks (fire episodes). Anthropogenic pollen indicators (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) reveal a significant increase between ca. 7500 and ca. 4500 years BP, with repeated peaks reaching ~700–1000 units, reflecting intensified human activity, including forest clearance, grazing, and the early expansion of agriculture. Cultivated tree pollen (*Castanea*, *Olea*, and *Pistacia*) reappears repeatedly during this period, with low but recurrent occurrences around 8500–8000, 7200, and 6500–6000 years BP, providing the earliest clear evidence of arboriculture and managed tree cultivation. While human impact becomes increasingly detectable, charcoal values suggest that burning was jointly controlled by climate, fuel availability, and the emergence of anthropogenic fire use, rather than being solely driven by human activity. The high peaks in Lake Eski Acıgöl and Lake Van also indicated human influence on nature, whereas the low fluctuations in the graphs of samples from the Balkans and Caucasus

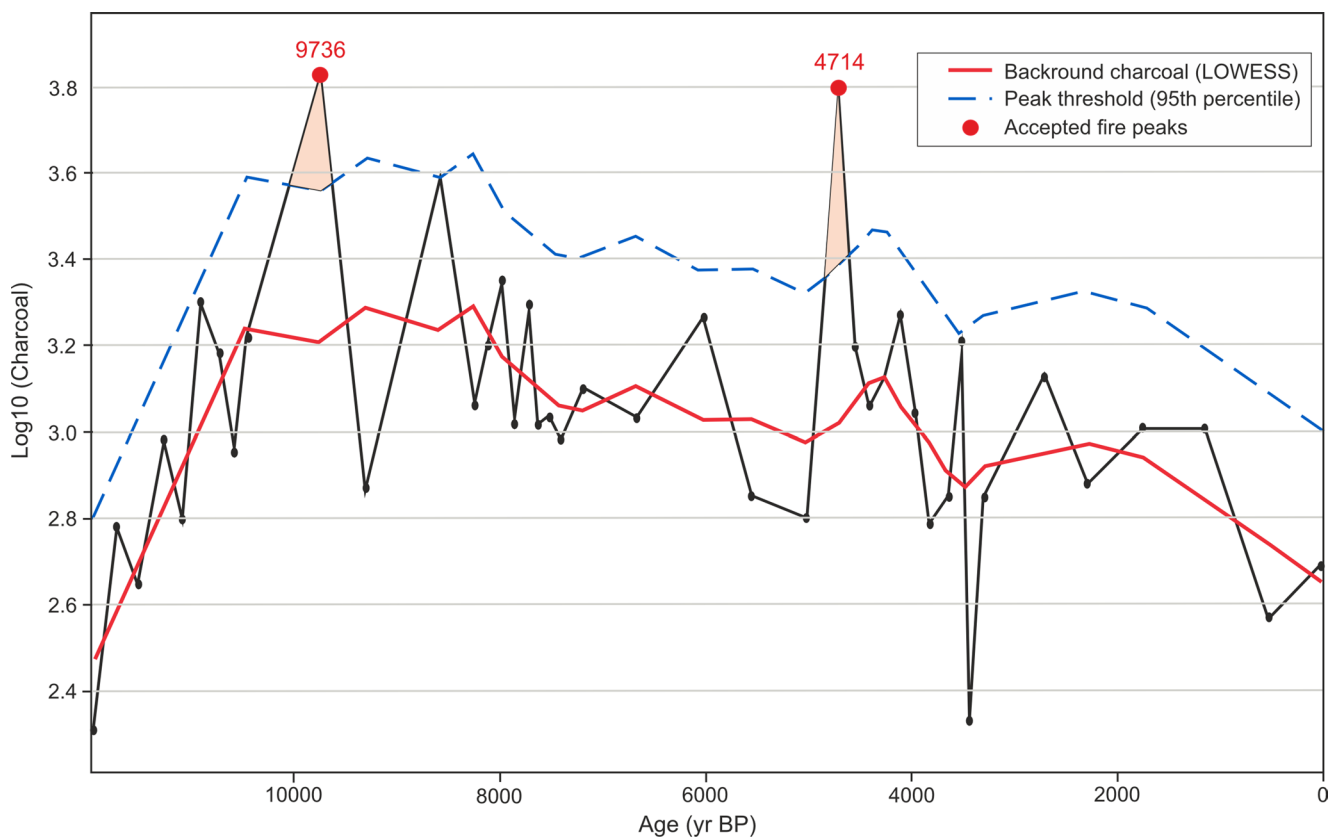


Fig. 6 Charcoal diagram showing $\log_{10}$ -transformed charcoal concentrations plotted against age (years BP), with the smoothed background curve and identified fire peaks. Black dots show individual measurements, and the black line shows the raw charcoal signal. The teal curve

represents background charcoal estimated using LOWESS smoothing. The dashed red line marks the 95th percentile threshold of the residual series, and shaded red areas indicate statistically significant fire peaks. Red markers denote accepted fire events

reflected a lower human influence (Fig. 7) (Connor et al. 2018; Brechbühl et al. 2024; Ganz et al. 2025). The strongest fire episode in the entire record occurs at around 4.7 ka BP, when the $\log$ -transformed charcoal count reaches nearly 3.9, and the total charcoal count is exceptionally high. Additional large peaks around 5.8–6.0 and 4.2 ka BP indicate repeated high-magnitude charcoal peaks (fire episodes). This period marks the main phase of interaction among optimal climatic conditions, maximum forest biomass, and increasing human disturbance. The intertwined relationship among human activity, climate, and high biomass levels resulted in the Holocene maximum in fire activity and the peak magnitude of charcoal. At the same time, deciduous *Quercus* percentages begin to decline after ca. 4500–4000 years BP, reaching moderate values of 20%, indicating increased forest opening and a more heterogeneous woodland structure.

5.1.3 Late mid-holocene to early late Holocene (ca. 4500–3000 years BP): increasing human control and changing vegetation structure

After ca. 4500 years BP, background charcoal values remain relatively high, though they display increasing variability. Meanwhile, peak magnitudes gradually decline. Several distinct peaks persist between 4000 and 3500 years BP, but overall charcoal concentrations decrease, indicating continued, albeit less extreme, fire activity. This phase likely reflects increasing climatic variability, combined with evolving land use strategies.

Deciduous *Quercus* begins to decline after ca. 4500–4000 years BP, signaling partial forest opening, fragmentation, and shifts in woodland composition. Anthropogenic pollen indicators (Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) remain elevated, and cultivated tree pollen becomes more frequent, suggesting the continued expansion of managed tree crops and more permanent agricultural systems. Fire regimes during this period appear increasingly shaped by human land

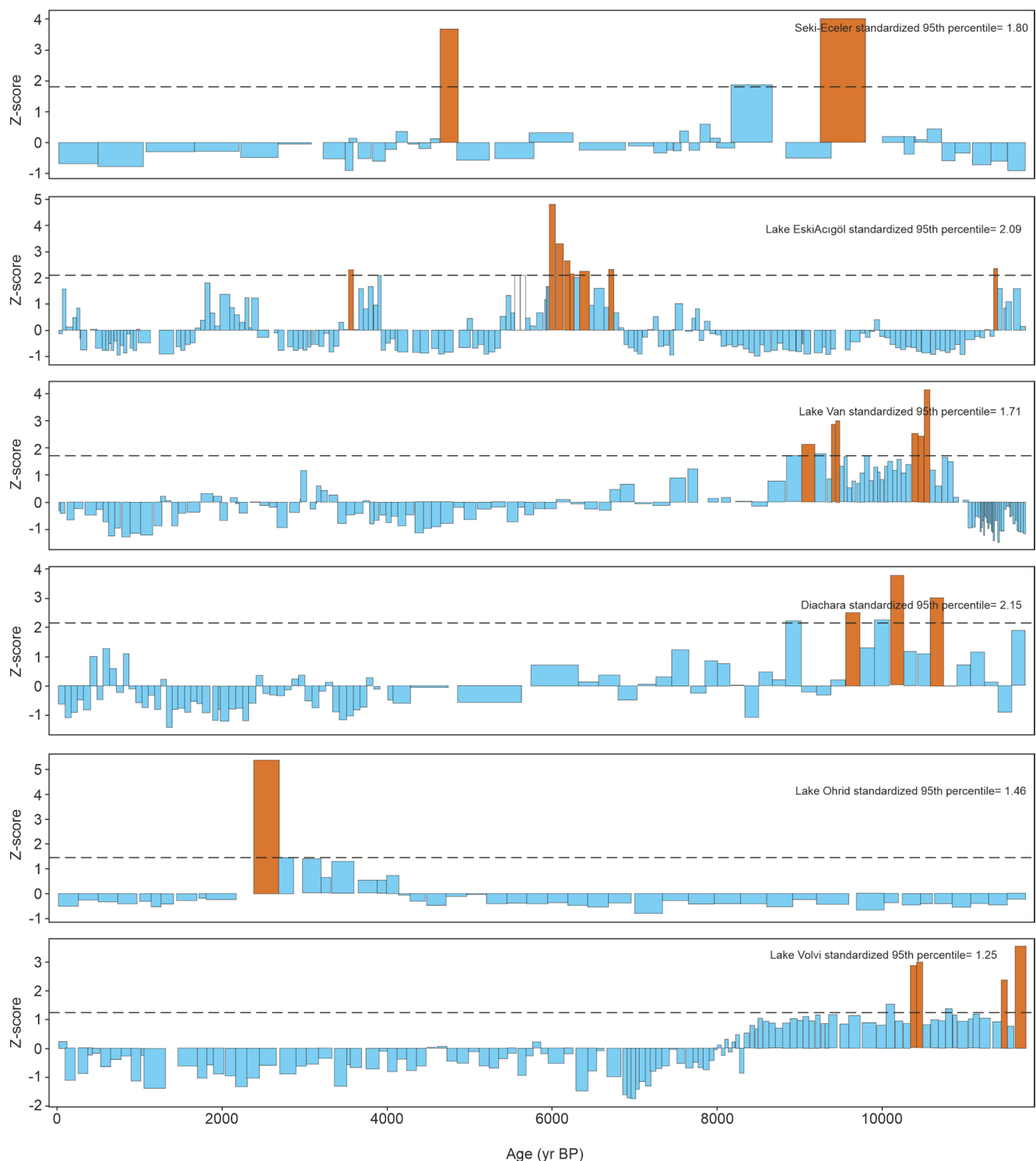


Fig. 7 Standardized charcoal records (z-scores) plotted against age (years BP) for the Seki-Eceler record (this study) and selected regional sites from the eastern Mediterranean and adjacent regions, including Lake Eski Acıgöl (Turner et al. 2008), Lake Van (Wick et al. 2003), Didachara (Connor et al. 2018), Lake Ohrid (Brechtbühl et al. 2024), and Lake Volvi (Ganz et al. 2025). The charcoal data are presented as bar diagrams to emphasize temporal resolution and sampling gaps.

Dashed horizontal lines indicate the 95th percentile threshold of each standardized record, while orange bars highlight values that exceed this threshold and are interpreted as significant fire episodes. Fire events are defined as peaks exceeding the 95th percentile threshold of the residual charcoal series. Peaks below this threshold are not considered significant fire events

use, with frequent, lower fire activity associated with clearance, grazing, and cultivation.

5.1.4 Late Holocene (ca. 3000 years BP–present): intensification of managed landscapes and decline of large-scale burning

A pronounced long-term decline in fire activity is characteristic of the Late Holocene. Log-transformed charcoal values decrease progressively to ~ 2.3 – 2.7 , and total charcoal counts mostly fall below ~ 4000 – 5000 , with fewer and weaker peaks. Fire activity decreased substantially relative to the Middle Holocene maximum, indicating a shift towards less fire-prone conditions.

The percentage of deciduous *Quercus* decreases to moderate levels (20–40%), reflecting more open and heterogeneous woodland landscapes. Anthropogenic pollen indicators (Asteraceae, Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago*) remain present and increase locally between ca. 4000 and 2500 years BP, suggesting sustained or intensified human occupation. Cultivated tree pollen becomes more prevalent, peaking around 3500–3000 years BP and again near 2000 years BP, indicating the consolidation of arboriculture and perennial cropping systems. Arboriculture and long-term perennial cropping practices had become firmly established at 2000 years BP. The combination of reduced charcoal accumulation and increased cultivation of trees (*Castanea*, *Olea*, and *Pistacia*) indicates a fundamental shift from fire-driven land clearance to more stable, intensively managed agricultural landscapes. Similar Late Holocene trends have been widely reported across the eastern Mediterranean. They are commonly associated with increased land use, greater grazing pressure, fragmented landscapes, and, more recently, the adoption of controlled burning and active fire suppression. Meanwhile, deciduous *Quercus* percentages remain moderate, anthropogenic pollen indicators persist, and cultivated tree pollen (*Castanea*, *Olea*, and *Pistacia*) shows prominent peaks around 3500–3000 years BP and again near 2000 years BP.

In summary, fire regimes in southwestern Anatolia were not controlled by a single driver, but rather emerged from the interplay of climate, fuel availability, forest structure, and human land use. Looking ahead, a better understanding of these dynamics could be achieved through sub-millennial charcoal analyses combined with quantitative fire-return interval modeling, as well as closer integration with archaeological datasets to better constrain the timing and nature of human fire use.

6 Conclusions

This study presents the dynamics of the Holocene fire regime in southwestern Anatolia using a high-resolution charcoal record alongside pollen-based indicators of vegetation and human activity. The main objective is to understand how fire activity responded to long-term changes in climate, vegetation structure and human land use. The analysis is based on palynological and charcoal data from the Holocene period. This includes a quantitative analysis of charcoal concentrations, estimation of background trends using LOWESS smoothing and detection of statistically significant fire peaks using a 95th percentile threshold. These datasets are evaluated alongside vegetation (deciduous *Quercus*) and anthropogenic pollen indicators in order to reconstruct interactions between fire, the environment and humans.

The results show that Holocene fire regimes were highly dynamic, being controlled by multiple interacting factors rather than a single driver:

- 1) During the Early Holocene (12,000–8500 years BP), fire activity was low and infrequent due to limited biomass availability, despite climatic warming. There was a gradual increase in fire activity that corresponded with postglacial vegetation expansion, which culminated in a statistically significant fire peak at around 9736 years BP.
- 2) During the Middle Holocene (8500–4500 years BP), fire activity intensified markedly, reaching its maximum during this period. This phase was characterised by the development of dense oak forests, increased fuel availability and rising anthropogenic influence. The most significant fire event occurred around 4714 years BP and reflects the combined effects of high biomass, climatic variability and human activity.
- 3) After 4500 years BP, fire regimes became more variable but generally less intense. This period marks a transition towards greater human control over landscapes, with continued, albeit lower-magnitude, fire activity linked to land use practices such as grazing and clearance.
- 4) During the Late Holocene (3000 years BP to the present), fire activity declined significantly. This coincided with the expansion of managed landscapes and increased arboriculture (*Castanea*, *Olea* and *Pistacia*), as well as a shift from natural fire regimes to more controlled or suppressed burning practices. The Holocene fire regimes in southwestern Anatolia were not driven by a single factor, but rather resulted from an evolving interplay between climate, vegetation structure and human activity. There was a clear long-term shift from climate-controlled to human-modified fire systems. Looking to the future, higher-resolution charcoal analyses, quantitative

fire-return interval modelling and closer integration with archaeological datasets would help to better constrain the timing, intensity and drivers of anthropogenic fire use.

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Author contributions ZH handled data processing and manuscript drafting. DB handled data processing and manuscript drafting. AMG provided archaeological expertise on Seki-Eceler Höyük and the regional historical context. SAÖ contributed to data visualisation and reviewed the draft. ZBÖ contributed to data visualisation and draft review. KZ contributed data visualisation. ÖY contributed to data visualisation and reviewed the draft.

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Data availability The data that support the findings of this study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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